

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

79%

QUESTIONS: 5

Total Marks: 50

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I S. Jeyaraj (192525348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Jeyaraj

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

	Problem Identification (20%)	Debugging (25%)	Optimization (20%)	Analysis (20%)	Code quality (15%)
Q1 9.5	2	2	2	2	1.5
Q2 9	2	2	2	2	1
Q3 8	1.5	2.5	1.5	1	1.5
Q4 7.5	1.5	2	1.5	1.5	1
Q5 5.5	1	1	1	1	1.5

39.5

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding(2)	Identifies most errors with minor gaps(1.5)	Identifies some errors but misses key issues(1)	Unable to identify errors correctly(0)
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues(2.5)	Uses appropriate debugging methods with minor errors(2)	Limited debugging approach; requires guidance(1.5)	Unable to debug or resolve issues
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Code Quality	15%	Produces clean, well-structured code with proper comments and documentation(1.5)	Code is mostly clear with minor documentation gaps(1)	Code lacks structure and proper documentation(0.5)	Poorly written code with no documentation

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Debugging & Optimisation Task

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I shaik.imran(192525348)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

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Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:
- **IP Address:** 192.168.10.65
 - **Subnet Mask:** 255.255.255.192 (/26)
 - **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

Network Configuration Debugging

Given Configuration

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

1. Identify the Subnet of the Configured IP

A /26 subnet mask provides blocks of **64 addresses**.

The subnets are:

1. 192.168.10.0/26 → 0–63
2. 192.168.10.64/26 → 64–127
3. 192.168.10.128/26 → 128–191
4. 192.168.10.192/26 → 192–255

Therefore, the IP address **192.168.10.65** belongs to:

Subnet: 192.168.10.64/26

2. Network Address, Broadcast Address and Host Range

Description	Address
Network Address	192.168.10.64
First Valid Host	192.168.10.65
Last Valid Host	192.168.10.126
Broadcast Address	192.168.10.127

3. Verify the IP Address and Gateway

IP Address: 192.168.10.65

The IP address **192.168.10.65** is a **valid host address** in the subnet **192.168.10.64/26**.

Default Gateway: 192.168.10.1

The gateway **192.168.10.1** belongs to the subnet:

192.168.10.0/26

Therefore, the workstation and default gateway are in **different subnets**.

4. Configuration Error

The main error is the **default gateway configuration**.

- Workstation subnet: **192.168.10.64/26**
- Configured gateway subnet: **192.168.10.0/26**

A default gateway must normally be configured with an IP address in the **same subnet as the workstation**.

5. Corrected Configuration

If the workstation needs to communicate with systems in **192.168.10.0/26**, an appropriate configuration is:

Setting	Correct Value
IP Address	192.168.10.2
Subnet Mask	255.255.255.192 (/26)
Default Gateway	192.168.10.1

Corrected Subnet Details

- **Network Address:** 192.168.10.0
- **Broadcast Address:** 192.168.10.63
- **Valid Host Range:** 192.168.10.1 – 192.168.10.62

The address **192.168.10.1** can be used as the gateway, and **192.168.10.2** can be assigned to the workstation.

6. Total Number of Usable Host Addresses

For a **/26** network:

- Total addresses = **$2^{(32 - 26)} = 64$**
- Network address = **1**
- Broadcast address = **1**

Usable Host Addresses

$64 - 2 = 62$ usable host addresses

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

IPv6 Network Configuration Debugging

Given IPv6 Addresses

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

1. Expand Both IPv6 Addresses

Device A

Compressed address:

2001:db8::ff00:42:8329

Complete notation:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

Compressed address:

2001:db8:0:0:1::1

Complete notation:

2001:0db8:0000:0000:0001:0000:0000:0001

2. Determine the Network Prefix

Since both addresses use /64, the first 64 bits represent the network portion.

Device A Network Prefix

2001:db8:0:0::/64

or in full notation:**2001:0db8:0000:0000::/64**

Device B Network Prefix

2001:db8:0:0::/64

The 0001 in Device B is after the first 64 bits, so it is part of the **interface/host portion**, not the network prefix.

3. Do Both Devices Belong to the Same Subnet?

Yes. Both devices belong to:

2001:db8:0:0::/64

Therefore, based on the given **/64 prefix**, both devices are in the **same IPv6 subnet**.

4. Identify the Addressing or Prefix Issue

There is **no subnet mismatch** in the given addresses.

Device	IPv6 Address	Network Prefix
Device A	2001:db8::ff00:42:8329/64	2001:db8:0:0::/64
Device B	2001:db8:0:0:1::1/64	2001:db8:0:0::/64

Both are correctly configured in the same /64 network.

If communication still fails, the problem may be due to another configuration issue such as:

- Incorrect IPv6 gateway
- Network interface disabled
- Firewall blocking communication
- Wrong VLAN or network connection
- Duplicate address or routing problem

5. Optimized IPv6 Configuration

A simple corrected configuration can be:

Device A

IPv6 Address: 2001:db8:0:0::1/64

Device B

IPv6 Address: 2001:db8:0:0::2/64

Common Network Prefix

2001:db8:0:0::/64

These addresses are easy to manage and clearly belong to the same subnet.

6. Number of Available Host Addresses

An IPv6 /64 network has:

- **128 total bits**
- **64 bits for network**
- **64 bits for host/interface**

Therefore:

Number of possible host addresses = 2⁶⁴

= 18,446,744,073,709,551,616 addresses

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

IPv6 Network Configuration Debugging

Given IPv6 Addresses

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

1. Expand Both IPv6 Addresses

Device A

Compressed address:

2001:db8::ff00:42:8329

Complete notation:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

Compressed address:

2001:db8:0:0:1::1

Complete notation:

2001:0db8:0000:0000:0001:0000:0000:0001

2. Determine the Network Prefix

Since both addresses use **/64**, the first 64 bits represent the network portion.

Device A Network Prefix

2001:db8:0:0::/64

or in full notation:

2001:0db8:0000:0000::/64

Device B Network Prefix

2001:db8:0:0::/64

The 0001 in Device B is after the first 64 bits, so it is part of the **interface/host portion**, not the network prefix.

3. Do Both Devices Belong to the Same Subnet?

Yes. Both devices belong to:

2001:db8:0:0::/64

Therefore, based on the given /64 **prefix**, both devices are in the **same IPv6 subnet**.

4. Identify the Addressing or Prefix Issue

There is **no subnet mismatch** in the given addresses.

Device	IPv6 Address	Network Prefix
Device A	2001:db8::ff00:42:8329/64	2001:db8:0:0::/64
Device B	2001:db8:0:0:1::1/64	2001:db8:0:0::/64

Both are correctly configured in the same /64 network.

If communication still fails, the problem may be due to another configuration issue such as:

- Incorrect IPv6 gateway
- Network interface disabled
- Firewall blocking communication
- Wrong VLAN or network connection
- Duplicate address or routing problem

5. Optimized IPv6 Configuration

A simple corrected configuration can be:

Device A

IPv6 Address: 2001:db8:0:0::1/64

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IPv6 Address: 2001:db8:0:0::2/64

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These addresses are easy to manage and clearly belong to the same subnet.

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An IPv6 /64 network has:

- **128 total bits**
- **64 bits for network**
- **64 bits for host/interface**

Therefore:

Number of possible host addresses = 2⁶⁴

= 18,446,744,073,709,551,616 addresses

Final Answer

Both Device A and Device B are already in the same subnet, **2001:db8:0:0::/64**, so there is no actual prefix mismatch. For easier network management, they can be assigned **2001:db8:0:0::1/64** and **2001:db8:0:0::2/64**. The corrected /64 subnet provides **2⁶⁴ possible interface addresses**.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.

Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements

Given Network

192.168.1.0/24

The organization uses subnet masks:

- /26
- /27
- /28

Since the exact number of hosts required by each department is not given, the following analysis shows the capacity of each subnet type and an example of an optimized **VLSM allocation**.

1. Subnet Details

A. /26 Subnet

Subnet mask: **255.255.255.192**

Each subnet contains:

- Total addresses = **64**
- Usable host addresses = **62**

Example:

Network Address	Host Range	Broadcast Address
192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63

B. /27 Subnet

Subnet mask: **255.255.255.224**

Each subnet contains:

- Total addresses = **32**
- Usable host addresses = **30**

Example:

Network Address	Host Range	Broadcast Address
192.168.1.64/27	192.168.1.65 – 192.168.1.94	192.168.1.95

C. /28 Subnet

Subnet mask: **255.255.255.240**

Each subnet contains:

- Total addresses = **16**
- Usable host addresses = **14**

Example:

Network Address	Host Range	Broadcast Address
192.168.1.96/28	192.168.1.97 – 192.168.1.110	192.168.1.111

2. Number of Usable Addresses

Subnet Mask	Total Addresses	Usable Hosts
/26	64	62
/27	32	30
/28	16	14

3 Identifying. Inefficient Allocation

An inefficient allocation occurs when a department receives a subnet much larger than its actual requirement.

Example

If a department requires only **10 hosts** but receives a /26 subnet:

- Available usable hosts = **62**
- Required hosts = **10**
- Unused addresses = **62 – 10 = 52**

This wastes **52 usable IP addresses**.

A better choice is a /28 subnet:

- Usable hosts = **14**
- Required hosts = **10**
- Unused addresses = **14 – 10 = 4**

Therefore, assigning /28 instead of /26 improves address utilization.

4. Possible Subnet Allocation Mistakes

The following mistakes can cause addressing problems:

1. **Overlapping subnets**

Two departments should not use overlapping address ranges.
2. **Incorrect subnet mask**

A device using /26 while other devices use /27 may cause communication problems.
3. **Using network or broadcast addresses**

These addresses cannot be assigned to normal hosts.
4. **Large subnet for a small department**

This causes unnecessary IP address wastage.
5. **Small subnet for a large department**

This causes insufficient IP addresses.

5. Optimized VLSM Allocation

VLSM allocates larger subnets to departments with more hosts and smaller subnets to departments with fewer hosts.

Example Department Requirements

Assume the organization has:

- Department A → **50 hosts**
- Department B → **25 hosts**
- Department C → **10 hosts**

Optimized Allocation

Department	Hosts Required	Subnet	Usable Hosts	Unused Hosts
Department A	50	192.168.1.0/26	62	12
Department B	25	192.168.1.64/27	30	5
Department C	10	192.168.1.96/28	14	4

6. Detailed VLSM Addressing Plan

Department	Network Address	Valid Host Range	Broadcast Address
A	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
B	192.168.1.64/27	192.168.1.65 – 192.168.1.94	192.168.1.95
C	192.168.1.96/28	192.168.1.97 – 192.168.1.110	192.168.1.111

The remaining address space:

192.168.1.112 – 192.168.1.255

can be reserved for future departments or network expansion.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.

- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

Given Addresses

Device A: `2001:db8::ff00:42:8329/64`

Device B: `2001:db8:0:0:1::1/64`

1. Expanded IPv6 Addresses

Device A:

`2001:0db8:0000:0000:0000:ff00:0042:8329`

Device B:

`2001:0db8:0000:0000:0001:0000:0000:0001`

2. Network Prefix

Since both devices use `/64`, the first 64 bits represent the network part.

Device A Network Prefix:

`2001:db8:0:0::/64`

Device B Network Prefix:

`2001:db8:0:0::/64`

3. Same Subnet Verification

Yes. Both devices belong to the same subnet:

`2001:db8:0:0::/64`

Therefore, there is **no IPv6 prefix mismatch** between the two devices.

4. Possible Communication Problem

The IPv6 addresses are valid and belong to the same `/64` subnet. If communication fails, possible problems may include:

- Incorrect IPv6 default gateway
- Wrong VLAN configuration
- Firewall blocking communication
- Network interface disabled
- Routing or switch configuration problem

5. Optimized IPv6 Configuration

For simple and easy management, configure:

Device A:

`2001:db8:0:0::1/64`

Device B:

`2001:db8:0:0::2/64`

Network Prefix:

2001:db8:0:0::/64

Both devices will clearly belong to the same subnet.

6. Number of Available Host Addresses

For an IPv6 /64 network:

Host bits = 128 - 64 = 64

Total possible host addresses:

2^64 = 18,446,744,073,709,551,616

Conclusion

Both original devices already belong to the same subnet, 2001:db8:0:0::/64, so there is no addressing or prefix mismatch. The communication failure is likely caused by another network configuration problem. An optimized configuration is Device A as 2001:db8:0:0::1/64 and Device B as 2001:db8:0:0::2/64. The /64 subnet provides **2^64 possible host addresses**.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Given Routing Table

192.168.1.0/24 → 10.0.0.2
192.168.2.0/24 → 10.0.0.3
Default Route → 10.0.0.1

Destination IP: **192.168.2.100**

1. Longest-Prefix Match

The destination IP **192.168.2.100** belongs to:

192.168.2.0/24

Therefore, the matching route is:

192.168.2.0/24 → 10.0.0.3

2. Next-Hop Router

The packet will be forwarded to:

10.0.0.3

3. Destination Subnet Verification

For the network **192.168.2.0/24**:

Network Address: **192.168.2.0**

Valid Host Range: **192.168.2.1 to 192.168.2.254**

Broadcast Address: **192.168.2.255**

The destination IP **192.168.2.100** is within this valid host range.

Therefore, the destination belongs to the correct subnet.

4. Possible Routing Errors

If the packet is not reaching the destination, possible errors are:

- Next-hop address **10.0.0.3** may be unreachable.
- The route may be incorrectly configured.
- The next router may not have a route to the destination.
- The destination device may be unavailable.
- There may be a wrong subnet mask or network configuration.

5. Optimized Routing Table

Destination Network: **192.168.1.0/24**

Next Hop: **10.0.0.2**

Destination Network: **192.168.2.0/24**

Next Hop: **10.0.0.3**

Default Route: **0.0.0.0/0**

Next Hop: **10.0.0.1**

6. Route Used When No Match Exists

If no specific route matches the destination IP address, the router uses the:

Default Route: 0.0.0.0/0 → 10.0.0.1

Conclusion

The destination **192.168.2.100** matches the network **192.168.2.0/24**. According to the longest-prefix match rule, the packet is forwarded to **10.0.0.3**. If no matching route is found, the router uses the default route and forwards the packet to **10.0.0.1**.

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Q2					
Q3					
Q4					
Q5					

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Code Quality	15%	Produces clean, well- structured code with proper comments and documentatio n(1.5)	Code is mostly clear with minor documentat ion gaps(1)	Code lacks structure and proper documentatio n(0.5)	Poorly written code with no documentatio n